

Polarization-Insensitive Optical Switches Based on Mode-Insensitive Devices

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Abstract—Owing to the waveguide birefringence in most photonic integrated circuits, optical switches are typically sensitive to the light state of polarization (SOP). In certain application scenarios, optical switches are required to process light signals with random SOPs from single-mode fibers. This study presents a method to achieve polarization-insensitive optical switches based on mode-insensitive devices after transforming the fundamental mode of one polarization component in randomly polarized light into a high-order mode of its orthogonal polarization, without extra power consumption or significant size increase. For 2×2 silicon-based thermo-optic switches, the measured polarization-dependent losses (PDLs) in both CROSS and BAR states were less than 0.4 dB at a wavelength of 1315 nm. Ranging from 1309 to 1339 nm, the PDLs were less than 0.9 dB. A 4×4 optical switch was also experimentally demonstrated, whose PDLs for all routing paths in all BAR and CROSS states were less than 1.2 dB at a wavelength of 1334 nm. This method can potentially be used to construct polarization-insensitive on-chip integrated photonic systems for applications in fields such as optical communications and optical interconnections.

Index Terms—Birefringence, optical switch, photonic integrated circuit, polarization independence.

I. INTRODUCTION

WITH rapid advancements in artificial intelligence, cloud computing, and other technologies, data traffic in data centers (DCs) and high-performance computers (HPCs) has surged [1], [2], [3]. As an important component of photonic integrated circuits (PICs), optical switches have the potential to replace traditional electrical switches in DCs and HPCs due to their low power consumption, low latency, and large bandwidth [4], [5], [6], [7], [8], [9], [10]. The cross-section size of PIC waveguides is typically restricted to a single-mode condition to achieve high-density integration, which leads to geometric birefringence. Additionally, the upper and lower claddings of the waveguide result in more complicated strain-birefringence effects. Both these factors determine the high sensitivity of PICs to the light state of polarization (SOP). In certain practical applications, optical switches are required to handle optical

signals from single-mode optical fibers, whose SOPs are usually not single-linear due to inevitable fiber bending and entanglement. Consequently, there is a need for polarization-insensitive switches to accommodate random SOP of input light.

Extensive studies have been conducted on polarization-insensitive optical switches or PICs. To reduce geometric birefringence under single-mode waveguide conditions, waveguides with axially symmetrical cross sections [11], [12] or subwavelength structures [13], [14], [15] have been proposed. However, these methods are doubly restricted by fabrication capabilities and internal stresses. Stress engineering can control the internal birefringence of a waveguide by supplying precise and consistent stress through stressed cladding, but it is difficult to control it accurately during fabrication [16], [17], [18], [19]. Another approach is the introduction of on-chip active polarization control systems before functional devices, which converts a random SOP into a single polarization via external modulations [20], [21], [22], [23]. However, this method induces extra power consumption and limits the simultaneous transmission of multiple-polarization optical signals. Moreover, the polarization-diversity scheme can effectively achieve polarization-insensitive functions [24], [25], [26], [27], [28]. The light SOP from the fiber can be decomposed into the combination of transverse electric (TE) and transverse magnetic (TM) polarizations in the waveguide. Orthogonally polarized components are typically split into two paths with the same TE or TM polarization by a polarization splitter and rotator (PSR). They are then processed using two sets of identical functional devices and finally combined into a single output port by another PSR. However, this scheme nearly doubles chip size and power consumption for the on-chip system. To address these issues, a non-duplicate polarization diversity scheme has been proposed, which transmits polarization components in separate paths within a single system [29], [30], [31]. However, only a limited number of networks satisfy these requirements. Recently, some researchers have introduced mode conversions both before and after the modulation section, which may not be suitable for scaling up the system [32].

In this study, we propose a scalable method for constructing polarization-insensitive optical switches, in which light with a random polarization is converted into different modes with the same polarization throughout a single mode-insensitive switch. First, one polarization component of a random light SOP is converted into a high-order mode of its orthogonal polarization component by using a polarization and mode converter (PMC). Subsequently, all the optical signals are processed using one set of mode-insensitive optical switch. Finally, the high-order

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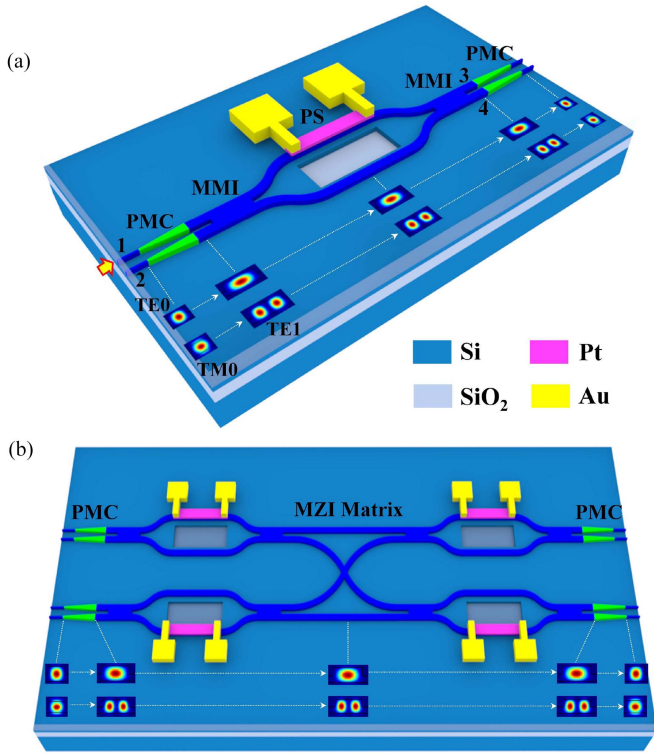


Fig. 1. Schematic of a polarization-insensitive (a) 2×2 optical switch and (b) 4×4 optical switch.

mode is converted back to its initial polarization by a reverse PMC and combined with the unchanged polarization component for output. Notably, this method can process optical signals with multiple polarizations simultaneously, without extra power consumption or significant size increase. Although the PMC is typically used as part of the PSR, it is used differently in our study. The silicon-based thermo-optic 2×2 switch was experimentally demonstrated with polarization-dependent losses (PDLs) of less than 0.4 dB at a wavelength of 1315 nm. Within the wavelength range of 1309 to 1339 nm, its PDLs and extinction ratios (ERs) were better than 0.9 dB and 19 dB respectively. In addition, a 4×4 switch was fabricated, and its PDLs for all routing paths at 1334 nm were less than 1.2 dB in all CROSS and BAR states. Ranging from 1323 to 1338 nm, the PDLs were less than 1.5 dB for half of routing paths in all BAR and CROSS states. The crosstalk for all paths in these two states was approximately -17 dB over a 44 nm bandwidth. This method can potentially be used to construct polarization-insensitive on-chip systems in fields such as optical communications and interconnections.

II. PRINCIPLE AND SIMULATIONS

A. Principle

The primary mechanism for achieving polarization-insensitive 2×2 and 4×4 optical switches is illustrated in Fig. 1(a) and (b), respectively. The 2×2 optical switch consists of two PMCs in both the input and output ports, and a mode-insensitive Mach-Zehnder interferometer (MZI)-type structure. The optical transmission process in the 2×2 switch

is divided into three stages. First, the TM0 component of input light at port 1 is converted into the TE1 light through the input PMC, whereas the TE0 component does not change, thus the orthogonal polarizations decomposed from a random SOP become a single polarization. Subsequently, the TE0 and TE1 lights are simultaneously processed by mode-insensitive switch: the TE0 and TE1 light beams are separately split into two light beams with the same amplitude and a difference of phase (DOP) of $\pi/2$ by the input multimode interference (MMI) coupler; when no voltage supply, they are merged by the output MMI coupler and subsequently output from port 4; when the DOP induced by the phase shifter (PS) is π , the two light beams interfere with each other and are output from port 3 of the output MMI coupler. Finally, the TE1 light is converted back to TM0 light through the output PMC and combined with TE0 light for the output. Therefore, for input lights with random polarizations, their TM- and TE-polarization components can be processed identically using only one set of devices or systems. In the 4×4 optical switch, a mode-insensitive MZI matrix is used to replace a single MZI structure of the 2×2 switch and the optical transmission process remains similar, as depicted in Fig. 1(b).

B. Simulation Results

All the devices described here were simulated for the O-band and were designed based on a silicon-on-insulator (SOI) substrate with a 340-nm-thick top silicon layer, 2- μm -thick buried oxide layer, and 1.5- μm -thick upper cladding. Although phase shifters with the strip waveguide can achieve a higher modulation efficiency, PMCs are typically achieved based on a ridge waveguide [33], [34], [35]. To simplify the fabrication, all the devices were designed based on ridge waveguides with a 190-nm-depth rid height. The simulation results were obtained using a commercial Lumerical tool.

1) *Polarization and Mode Converter*: Owing to the vertical asymmetry of the ridge waveguide, conversion from TM0 to TE1 may occur inside [33], [34], [35]. To determine the width of the PMC, the relationship between the effective refractive index of various modes and waveguide width was simulated using the Lumerical FDE engine, as depicted in Fig. 2(a). The conversion of TM0-TE1 occurred when the width of the waveguide was nearly 0.87 μm . Here, the width variation for a linear taper-shaped PMC was chosen from 0.8 to 0.95 μm , while the length of PMC was swept from 50 to 400 μm through the Lumerical FDTD simulation, as illustrated in Fig. 2(b). The conversion efficiency of TM0-TE1 (red line) was greater than 97% without TE0 light conversion when the length was greater than 90 μm . To increase the process tolerance and improve the conversion efficiency, a PMC length of 400 μm was chosen in the 2×2 switch. Under these parameters, the light propagation profiles at 1310 nm for the TE0 and TM0 lights are shown in the insets of Fig. 2(c) and (d), respectively. The red lines reveal that the insertion losses (ILs) of a single PMC for TE0 and TM0 lights were smaller than 0.05 dB over an 80 nm bandwidth. The crosstalk (CT) from the unconverted TM0 light for the TM0 light input was less than -19 dB, as depicted by the black line in Fig. 2(d).

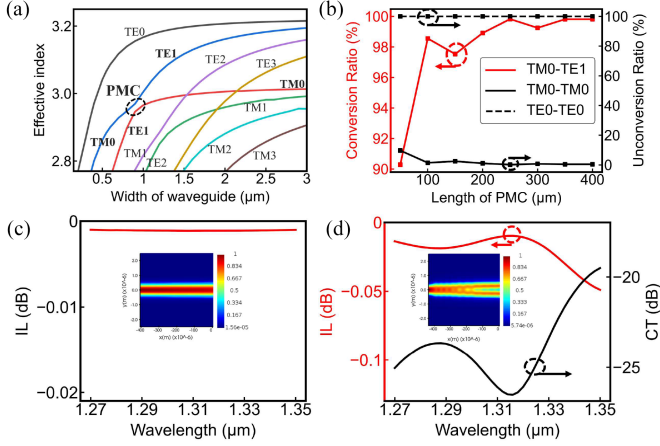


Fig. 2. (a) Effective refractive index of various modes vs. waveguide width. (b) Conversion ratio vs. length of polarization and mode converter (PMC) at 1310 nm. (c) Insertion loss (IL) vs. wavelength for TE0 light. (d) IL and crosstalk (CT) vs. wavelength for TM0 light input.

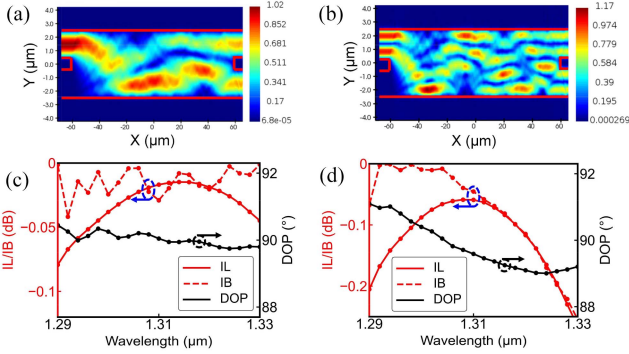


Fig. 3. (a)-(b) Optical fields of a mode-insensitive 2 × 2 MMI coupler for (a) TE0 and (b) TE1 light inputs at 1310 nm. (c)-(d) IL, power imbalance (IB), and difference of phase (DOP) between output ports vs. wavelength for (c) TE0 and (d) TE1 light inputs.

2) Mode-Insensitive 2 × 2 MMI Coupler: Owing to its large bandwidth and fabrication tolerance, a 2 × 2 MMI coupler is a common choice for achieving a 3 dB splitter. Here, the mechanism of general interference was adopted, in which the input light is evenly split into two output ports of the MMI coupler when the length of the MMI multimode area is approximately 1.5 × the beat length, quadratic to its equivalent width of the multimode area [36]. The 5-μm multimode area width was chosen to balance the distance between the output ports and the size of the device. After sweeping using the Lumerical FDTD simulation tool, the length of the multimode area was determined to be 121 μm. At the junction of the multimode area and input/output ports, a waveguide width of 2.2 μm was chosen to reduce mode mismatching. The propagation profiles at 1310 nm for the TE0 and TE1 mode lights are shown in Fig. 3(a) and (b), respectively. Fig. 3(c) and (d) demonstrate that within the wavelength range of 1.29 to 1.326 μm, the power imbalances (IBs; red dotted line) between output ports were smaller than 0.2 dB for TE0 and TE1 mode lights, and the ILs (red solid line) were smaller than 0.08 dB for TE0 mode light and 0.2 dB for TE1 mode light. In addition, the mode-dependent loss was smaller

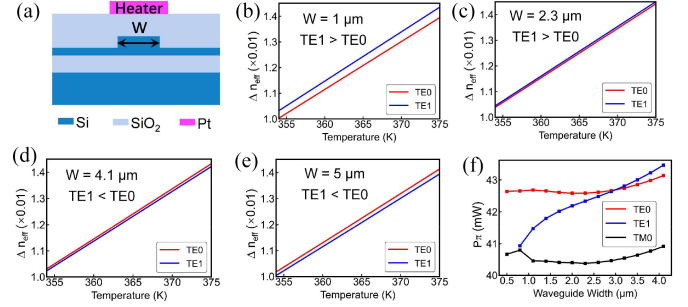


Fig. 4. (a) Schematic of the phase shifter. (b)-(e) Change in the effective index (Δn_{eff}) for TE0 and TE1 modes vs. temperature of the waveguide center at the waveguide width of (b) 1 μm, (c) 2.3 μm, (d) 4.1 μm, and (e) 5 μm. (f) Power consumption for phase change π (P_π) vs. waveguide width.

than 0.15 dB. Notably, the DOPs (the black line) between the two output ports fluctuated around the ideal 90 degrees, varying between +1.1 and −0.9 degrees for TE0 and TE1 mode lights. Therefore, for these two light modes, an MZI composed of a designed mode-insensitive MMI coupler can achieve the same switching function under the same phase shift.

3) Mode-Insensitive Phase Shifter: The phase shifters of the MZI switch are composed of multimode waveguides and thin-film heaters over the upper cladding, as depicted in Fig. 4(a). The width and thickness of the heater were fixed at 3 μm and 100 nm, respectively. According to the thermo-optic effect, the power consumption of the heater changes the waveguide temperature, thereby altering the refractive index of the silicon material and the mode effective refractive index. The phase change ($\Delta\varphi$) is calculated as $2\pi/\lambda \times \Delta n_{\text{eff}} \times L$, where Δn_{eff} , L , and λ denote the change in the mode effective index, modulation length, and operating wavelength, respectively. The values of L and λ are fixed at 50 μm and 1.31 μm. P_π is defined as power consumption for phase change π . Under the thermal influence, when the values of Δn_{eff} for the TE0 and TE1 modes are close, their values of $\Delta\varphi$ are consistent, thus achieving mode insensitivity and a similar P_π . Through a co-simulation of the Lumerical Heat module and FDE engine, we explore the impact of multimode waveguide width on mode independence, referring to the literature [37]. First, at waveguide widths of 1, 2.3, 4.1, and 5 μm, the relationship curves between Δn_{eff} and the waveguide temperature are simulated and depicted in Fig. 4(b)–(e), respectively. The value of Δn_{eff} increased linearly with the rising waveguide temperature. The relationship of Δn_{eff} for TE0 and TE1 lights changes with the widened waveguide width; compared to TE0 light, the value of Δn_{eff} for TE1 mode is separately larger, closer, and smaller as the waveguide widens from 1 to 5 μm, indicating that the mode-insensitive width exists. Next, we obtain the relationship curves between P_π and the waveguide width for TE0, TE1, and TM0 lights, as shown in Fig. 4(f). The width with the same P_π for the TE0 and TE1 modes is approximately 2.8 μm. In addition, the curve for the TM0 light also appears, which is distinctly different from that of the TE0 light. By balancing modulation efficiency and losses of bent waveguide connecting phase shifters and MMI couplers, a width of 2.3 μm was selected with only 0.2-mW P_π difference for TE0 and TE1 lights.

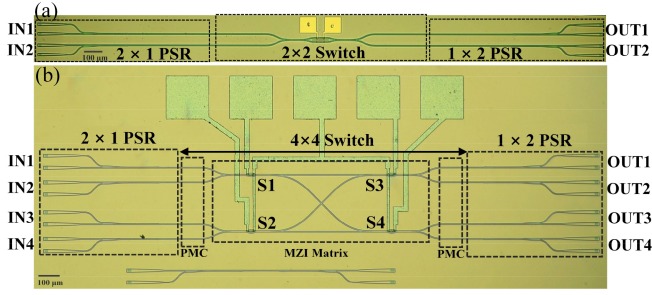


Fig. 5. Micrograph views of (a) a 2×2 switch and (b) a 4×4 switch.

III. FABRICATION AND MEASUREMENT RESULTS

A. Fabrication

To experimentally verify the designed devices, we fabricated 2×2 and 4×4 optical switches, as shown in Fig. 5(a) and (b), respectively. The devices were fabricated using an SOI chip with a 340-nm-thick top silicon layer. After electron beam lithography, the chip was etched by 190 nm using inductively coupled plasma etching. Subsequently, a $1.5\text{-}\mu\text{m}$ -thick silica cladding layer was deposited using plasma-enhanced chemical vapor deposition. The pattern of the heater was formed by direct-write laser (DWL) lithography, and a 100-nm-thick platinum layer was sputtered and peeled off to form a thin-film heater. After the gold electrode pattern was formed using DWL lithography, an 800-nm-thick gold layer was evaporated and peeled off to form the electrode. For low-loss coupling and convenient testing, grating couplers (GCs) were used to couple light into and out of the chip, which combined with PSRs to form a single TE0 or TM0 light as the input light of the switch and to receive TE0 and TM0 lights from the switch, explained in subsequent Section B. To reduce mode-dependent scattering loss for TE0 and TE1 modes influenced by waveguide sidewall roughness and avoid the excitation of higher-order modes as much as possible, the width of the bent waveguides connecting the switch units and waveguide crossing was set to $1.4\text{ }\mu\text{m}$ in the 4×4 optical switch.

B. Test Method for Polarization-Insensitive Switches

The polarization dependence can be evaluated by individually testing the transmissions of TE and TM polarizations, because the light SOP can be decomposed into a combination of TE and TM polarizations in the waveguide. Under an identical power consumption, if the same switching state can be obtained for a single TE and TM polarization light, the switching action will be achieved for their combination owing to the independent transmission of the TE and TM polarizations. To separately measure the transmission of TE and TM polarizations, a 2×1 PSR is added at the input of the switch to form quasi TE0 or TM0 lights, and a 1×2 PSR is added at the output of the switch to split and receive TE0 and TM0 light at different ports, as shown in Fig. 6. The TE GCs used in the switch only work for TE polarization. When TE0 light is coupled into port 1, it is converted into TM0 light through the 2×1 PSR and serves as the TM0 input light of the switch; The TM0 light from the switch is then converted into TE0 light through the 1×2 PSR

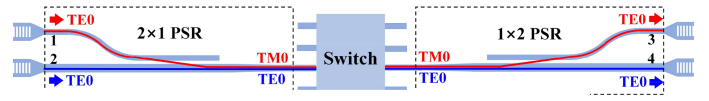


Fig. 6. Test method for single TE or TM light input.

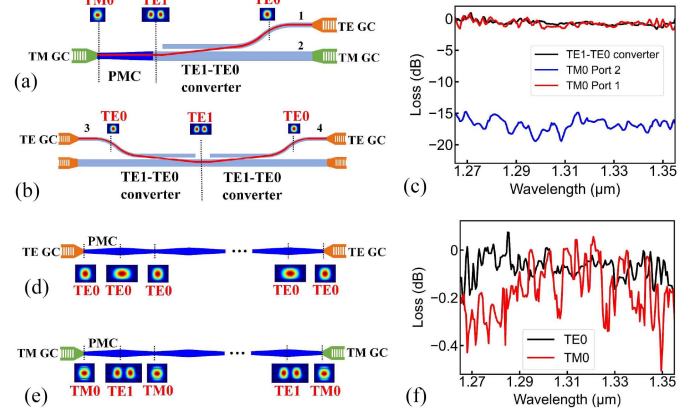


Fig. 7. (a) Test structure of PMC for TM0 light inputs. (b) Schematic of a pair of TE1-TE0 converters. (c) Measured results of overall test structure. (d)-(e) Schematic of cascading PMC for (d) TE0 and (e) TM0 light inputs. (f) Measured results of a single PMC through cascading.

and output from port 3. When the TE0 light is coupled into port 2, it remains unchanged after passing through the 2×1 PSR and serves as the TE0 input light of the switch; it passes through the switch and transmits to port 4 of the 1×2 PSR. By subtracting the transmission loss of a pair of 2×1 and 1×2 PSRs from the on-chip losses, the transmission losses of the TE0 and TM0 light in the switch are obtained.

For the optical path, the light was emitted from a tunable laser (Santec TSL-550) to a three-ring polarization controller (PC) and then coupled into and out of the optical chip. Finally, the light was received by an optical power meter (OPM; Yokogawa AQ2211). The fiber SOP was calibrated using a polarizing beam splitter cube and PC. For the electrical path, electrical signals from a precision voltage source (PVS; Keysight B2901BL) were supplied to the electrode using a probe. A computer program controls the PVS and receives OPM data.

C. Test of PMC

To evaluate the conversion efficiency of TM0-TE1 in PMC, we introduce an extra adiabatic TE1-TE0 converter [38] after PMC to convert TE1 light into TE0 light as direct measurement of TE1 light is difficult. The overall test structure for the PMC is shown in Fig. 7(a). TE GC was used for the coupling of TE polarization light; while TM GC was for the coupling of TM polarization light and was only used in this section. When TM-polarized light is coupled into the test structure through the TM GC, the light undergoes a TM0-TE1 conversion in the PMC and then a TE1-TE0 conversion in the converter; it is finally received by the TE GC at port 1. Efficient conversion of TM0-TE1 will be achieved by a PMC when all TM0 light is collected at port 1 and no TM0 light is measured at port 2. The measured results of the test structure in Fig. 7(a) are illustrated in Fig. 7(c). For TM0 light

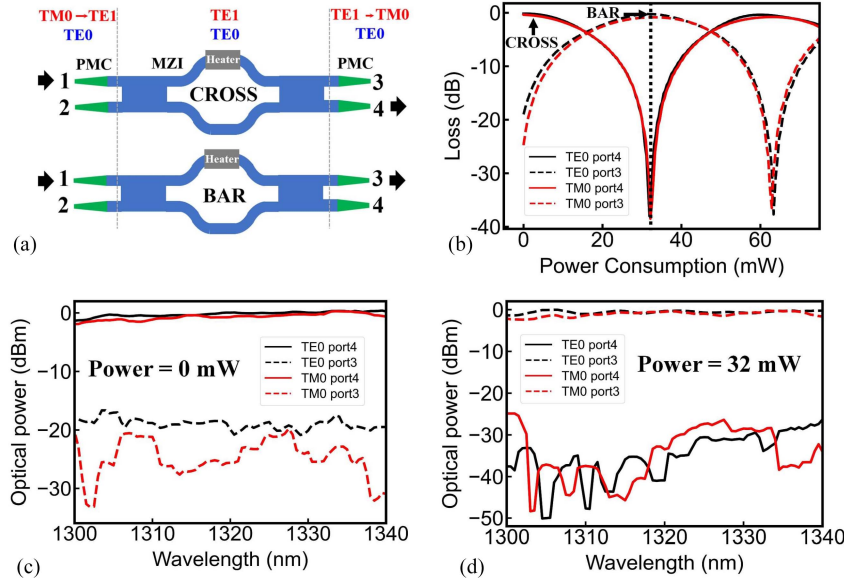


Fig. 8. (a) Schematic of the CROSS and BAR states. (b) Loss vs. power consumption at ports 4 and 3 for TM0 and TE0 lights input from port 1. (c)-(d) Optical power vs. wavelength in the (c) CROSS state and (d) BAR state.

input, the measured on-chip loss at port 2 (blue line) was larger than 15 dB, corresponding to a 3% residual of TM0 light and nearly 97% conversion. Meanwhile, the measured loss at port 1 (red line) was approximately 1.4 dB, primarily attributable to the insertion loss of a TE1–TE0 converter. To confirm it, two identical TE1–TE0 converters were connected to obtain the insertion loss during conversion, as shown in Fig. 7(b). The TE0 input light from port 3 went through a TE0–TE1 conversion in the first converter and a TE1–TE0 conversion in the second converter; it was finally output from port 4. Thus, the loss in the single TE1–TE0 converter was half of the on-chip loss for the system in Fig. 7(b). The test results for a single TE1–TE0 converter are represented by the black line in Fig. 7(c), which is almost consistent with the red line, indicating that efficient TM0–TE1 conversion occurs in the PMC.

Furthermore, 20 PMCs were cascaded in a back-to-back configuration to obtain an accurate loss of a single PMC for separate TE0 and TM0 light inputs, as depicted in Fig. 7(d) and (e). The calculated loss for a single PMC is shown in Fig. 7(f). Notably, over a 60 nm bandwidth from 1285 to 1345 nm, losses of a single PMC for TE0 and TM0 light inputs were in the range of 0–0.17 dB and 0–0.34 dB, respectively.

D. Test of the 2 × 2 Switch

The 2 × 2 switch was separately measured under quasi TE0 and TM0 light input. In Fig. 8(a), it is evident that the light from port 1 is output to port 4 in the CROSS state of the 2 × 2 switch, whereas the light from port 1 is output from port 3 in the BAR state. Fig. 8(b) depicts the measured curves between power consumption and optical power at a wavelength of 1315 nm when light is input from port 1. The solid and dotted black curves are the test results of the TE0 light input at ports 4 and 3; the solid and dotted red curves are the test results of TM0 light at ports 4 and 3. Notably, the curves at the same output

port coincide completely for the two polarizations. When no power was consumed, the switch was in the CROSS state. The loss of the 2 × 2 switch was smaller than 0.5 dB for TE0 and TM0 lights, and the PDL was approximately 0.3 dB. When the applied power consumption was approximately 32 mW, the device was switched to the BAR state. The losses on the chip were 0.3 dB and 0.7 dB separately for TE0 and TM0 lights. Fig. 8(c) and (d) illustrate wavelength-dependent properties in the CROSS and BAR states. Ranging from 1309 to 1339 nm, the PDLs were smaller than 0.9 dB, and the ERs were larger than 19 dB. The average PDL was smaller than 0.75 dB over a 60 nm bandwidth. Therefore, a polarization-insensitive 2 × 2 switch was experimentally demonstrated. Design optimizations of the MMI coupler and multimode bent waveguides can further reduce PDLs. In addition, the larger waveguide propagation loss for the TE1 mode is part of the PDL. The optical profile of the TE1 mode was closer to the waveguide sidewall than that of the TE0 mode, as shown in Fig. 1. Owing to the poor etching effect, the sidewall exhibited periodic fluctuations, resulting in a larger scattering loss for TE1 light.

E. Test of the 4 × 4 Switch

A 4 × 4 butterfly-type switch was fabricated based on 2 × 2 switches, as shown in Fig. 5(b). In the 4 × 4 switch, the PMC length was set to 100 μm, achieving a TM0–TE1 conversion efficiency of over 98.6%, as shown in Fig. 2(b). The TE0 and TM0 lights formed by the input 2 × 1 PSR were individually transmitted to the input PMC. While TE0 light did not change, TM0 light was converted into TE1 light. Subsequently, both TE0 and TE1 lights entered the MZI matrix. Through the output PMC, the TE0 light remained unchanged, but the TE1 light returned to the TM0 light. They were received and split by the output 1 × 2 PSR. The input and output ports of the 4 × 4 switch are labeled IN_i (i = 12,34) port and OUT_j (j = 12,34)

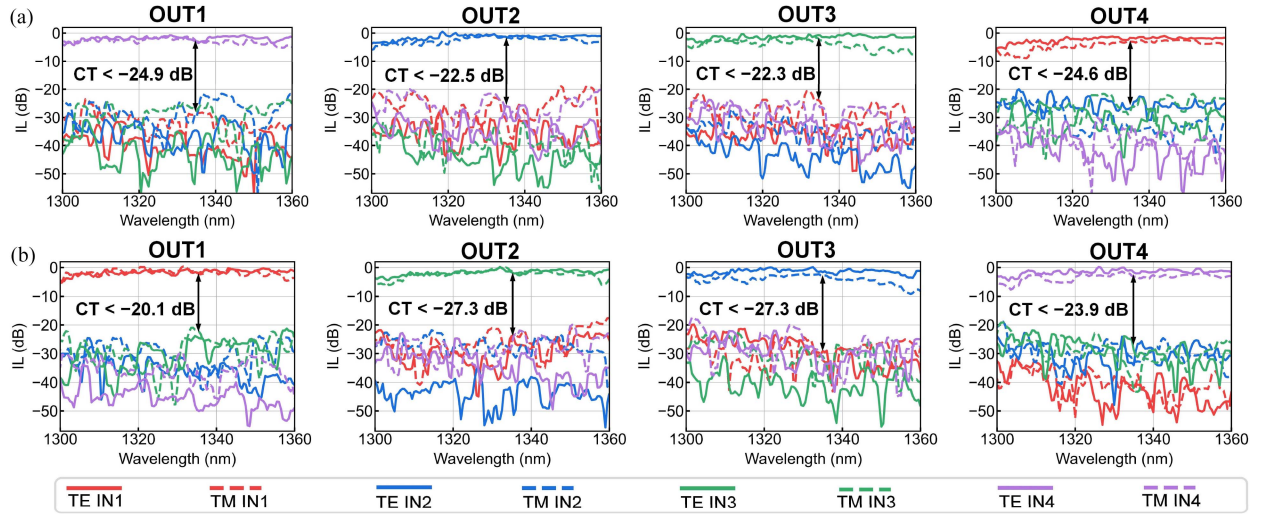


Fig. 9. Test curves for the 4×4 switch in (a) all CROSS state and (b) all BAR state.

TABLE I
PERFORMANCES OF THE 4×4 SWITCH AT A WAVELENGTH OF 1334 NM

State	Routing Path	IL _{TE} (dB)	IL _{TM} (dB)	PDL (dB)	CT _{TE} (dB)	CT _{TM} (dB)
All CROSS	IN1-OUT4	2.24	2.98	0.74	-24.63	-24.73
	IN2-OUT2	1.53	1.15	0.38	-28.7	-22.57
	IN3-OUT3	0.85	1.53	0.68	-30.26	-22.36
	IN4-OUT1	2.44	1.70	0.74	-32.73	-24.96
All BAR	IN1-OUT1	1.89	1.28	0.61	-22.79	-20.12
	IN2-OUT3	1.51	2.32	0.81	-29.55	-27.36
	IN3-OUT2	1.53	0.95	0.58	-29.01	-27.38
	IN4-OUT4	0.96	2.08	1.12	-23.92	-28.75

port, respectively. All BAR and all CROSS states are defined as situations in which all units are in the BAR or CROSS states, respectively. Fig. 9 illustrates the performances in these two extreme states. In each plot of Fig. 9, the red, blue, green, and purple curves represent test results for light input from IN1-OUT4 ports, respectively, at a fixed output port; with the solid and dotted lines representing the results of the TE₀ and TM₀ light inputs. Crosstalk of the routing path IN_i-OUT_j is defined as the ratio of maximum leaked output power from other input ports IN_m ($m \neq i$), $P_{\text{out}}(\text{IN}_m\text{-OUT}_j)$, to the ideal output power, $P_{\text{out}}(\text{IN}_i\text{-OUT}_j)$.

A group of voltages was supplied to the units to achieve all CROSS states for both polarizations. As shown in Fig. 9(a), the routing optical paths of IN1-OUT4, IN2-OUT2, IN3-OUT3, and IN4-OUT1 were achieved when light was input from the IN1, IN2, IN3, and IN4 ports, respectively. Overall performances at the wavelength of 1334 nm were summarized in Table I. The PDLs and the crosstalk of all these paths for both polarizations were less than 0.8 dB and -20 dB at 1334 nm. Over a 44 nm bandwidth from 1312 to 1356 nm, the crosstalk was approximately -17 dB. Furthermore, to simultaneously achieve all BAR state for both polarizations, the driving voltages of the four units were separately set to the corresponding values of the

BAR state. When light was input from the IN1, IN2, IN3, and IN4 ports, the routing optical paths of IN1-OUT1, IN2-OUT3, IN3-OUT2, and IN4-OUT4 were obtained, as illustrated in Fig. 9(b). The PDLs of these paths were less than 1.15 dB and the crosstalk was less than -20.7 dB at 1334 nm, comprehensively summarized in Table I. Over a 50 nm bandwidth, the crosstalk for these two polarizations was less than -17 dB. Moreover, ranging from 1323 to 1338 nm, the PDLs were less than 1.5 dB for routing paths of IN2-OUT2 and IN4-OUT1 in all CROSS state. Similarly, ranging from 1321 to 1345.5 nm, the PDLs were less than 1.5 dB for routing paths of IN1-OUT1 and IN3-OUT2 in all BAR state. Lower PDLs were obtained for these paths than those for other paths in the same state, because they were all transmitted through the upper-right switch unit S3 in Fig. 5(b), whereas other higher-PDL paths were transmitted through the lower-right unit S4. It can be inferred that unit S4 had a larger PDL, probably due to fabrication errors.

In the 4×4 switch, mode-dependent scattering loss for TE₀ and TE₁ lights can further be mitigated by optimizing the etching process or widening the bent waveguide slightly. In addition, the maximum write field size of the electron-beam lithography was $500 \mu\text{m} \times 500 \mu\text{m}$, and as the length of the device increased, the splicing error of the write field accumulated. It also has a significant impact on the transmission of the TE₁ light. These issues will be addressed in future research.

IV. CONCLUSION

In this study, we propose a method to achieve polarization-insensitivity optical switches by transforming one polarization component of random light into a high-order mode of its orthogonal polarization, without additional power consumption or significant size increase. The polarization-dependent loss of the fabricated silicon-based thermo-optic 2×2 switch was less than 0.9 dB at 1309-1339 nm, and the extinction ratio was larger than 19 dB for both polarizations. The polarization-insensitive switching function was effectively demonstrated in a 4×4 switch, with its polarization-dependent loss below 1.2 dB at

1334 nm. In all BAR and CROSS states, the polarization-dependent losses were less than 1.5 dB for half of routing paths ranging from 1323 to 1338 nm, and the crosstalk for all routing paths was approximately -17 dB over a 44 nm bandwidth. Owing to the widened bent waveguide, there is a possibility of excitation of unwanted higher modes and a larger footprint, which could be further optimized. Similarly, polarization-insensitive optical switches can be constructed on the 220-nm silicon-on-insulator platform and other photonic integrated platforms. Additionally, this approach also shows potential for the design of polarization-insensitive on-chip photonic devices and systems (such as modulators, waveguide crossings, and microrings), simultaneously supporting optical signals with multiple polarizations.

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